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TEST REPORT

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Attention to : Mr Shu Changhong

Project Code/Title : Reclamation Sand Testing

Test Method : 1) Particle Size Distribution
2) Silt and Coral/Shell Content
3) Moisture Content
4) Shear Strength by Direct Shear (Small Shearbox Apparatus)
5) Determination of Organic Content
6) 12 Metallic Elements Analysis

Test Standards : 1) BS 812-103.1:1985
2) BS 812-103.1:1985 & SANS 5840:2008 (Ignition Method)
3) BS 1377-2:1990 Clause 3(Part 3.2)
4) BS 1377-7:1990 Clause 4(Part 4.5.4)
5) BS 1377-3: 2018 Section 4
6) EPA 3051A & 6010D

Job Ref. : **X-2026-522-2**
Vessel Name : **JIAHE 99**
VOY No. : **JH99-96N**
Client Ref./Sample ID : **3-A**
Sampling Date : **08/04/2026**
Date Received : **10/04/2026**
Date of Report : **13/04/2026**

Total Pages : 5 (including cover page)
Remarks : Please refer to Appendix for photographs of samples as received



LA-1988-0020-B
LA-2019-0727-A
LA-2022-0815-F

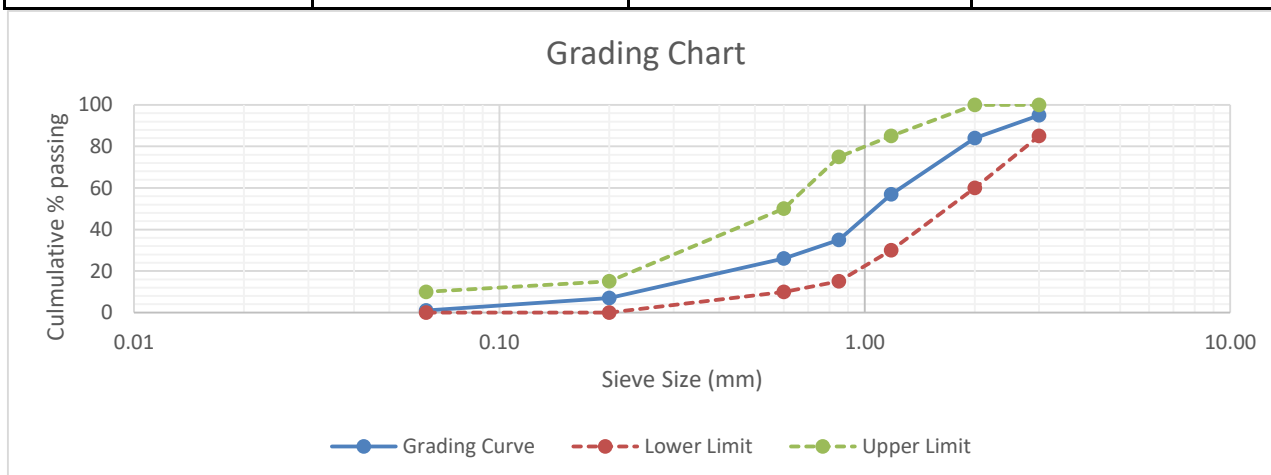
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1. Determination of Particle Size Distribution (BS 812-103.1:1985)

Sieve Size(mm)	Culmulative % Passing	Lower Limit (JTC Requirement)	Upper Limit (JTC Requirement)
3.00	95	85	100
2.00	84	60	100
1.18	57	30	85
0.850	35	15	75
0.600	26	10	50
0.200	7	0	15
0.063	1	0	10



Remarks: 1) Sand tested met JTC requirements for overall limits.
2) Washing and sieving method was used for the above test.

2. Silt Content (BS 812-103.1:1985) and Coral / Shell Content (SANS 5840:2008)

Silt Content (%) from passing 0.063mm sieve	0.5	JTC Requirement
Coral /Shell Content (%)	0.7	
Total (%)	1.2	Not more than 15%

3. Determination of Moisture Content (BS 1377-2:1990 Clause 3(Part 3.2))

Moisture Content (%)	9.7
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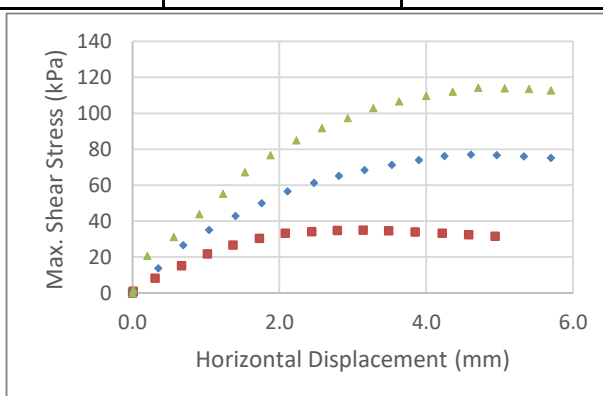
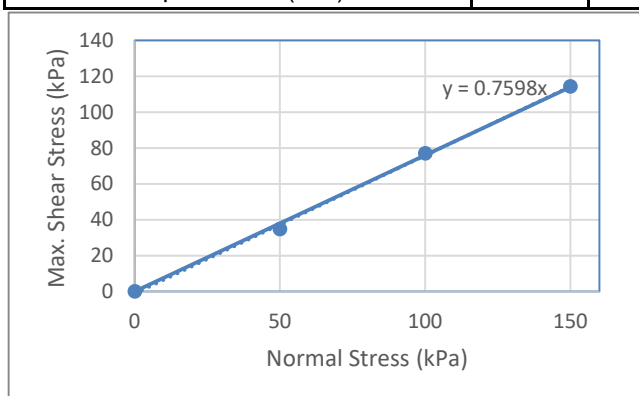
Remarks: Oven-drying method was used for the determination of moisture content



4. Shear Strength by Direct Shear (Small Shearbox Apparatus) (BS 1377-7:1990 Clause 4 (Part 4.5.4))

Maximum Dry Density, Mg/m ³		1.68
Minimum Dry Density, Mg/m ³		1.43
% Retained on 2.0mm Sieve (%)		16
Shearing Rate (mm/min)		1.5
Condition for relative density of 35%	Initial Bulk Density (Mg/m ³)	1.65
	Initial Dry Density (Mg/m ³)	1.51
Angle of Shearing Resistance ϕ' (°)		JTC Requirement
38		Limit of 32°-45°

Normal Stress (kPa)	0	50	100	150
Max. Shear Stress (kPa)	0	35	77	114
Horizontal displacement (mm)	0	3.14	4.61	4.71



5. Determination of Organic Matter Content (BS 1377-3: 2018, Section 4)

Organic Matter Content (%)	0.11
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6. 12 Metallic Analysis (EPA 3051A & EPA 6010C)

Element	Results (ppm)	Upper Limit Concentration(ppm)
Arsenic, As	N/A	30
Barium, Ba	N/A	200
Cadmium, Cd	N/A	2
Cobalt, Co	N/A	20
Chromium, Cr	N/A	100
Copper, Cu	N/A	35
Lead, Pb	N/A	100
Mercury, Hg	N/A	0.5
Molybdenum, Mo	N/A	10
Nickel, Ni	N/A	35
Selenium, Se	N/A	20
Zinc, Zn	N/A	200

Remarks:

- < = Less than
- Samples are digested according to EPA3051A, measured using ICP-OES
- All tested items are expressed in ppm, which equivalent to mg/kg

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APPENDIX

Photographs of sample received:

